

World as a Puzzling System: An Essay

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Abstract

Causal reasoning often begins with an arrow: X happens, and Y follows. In an evolving system, however, Y can change the conditions under which X occurs next. This essay asks what causal attribution requires when causes and consequences continue interacting. Beginning with the question of whether the times make the hero or the hero makes the times, we follow the discussion through feedback, stable states, response paths, intervention, and adaptation. Concrete examples show why different mechanisms can produce the same observed state and why following a trajectory can help distinguish them. We then examine how observers' predictions and explanations can influence the process they seek to understand, using the AI expansion as an unfolding case. A distinction between causal separability and causal salience keeps the evidence capable of distinguishing explanations separate from the pressure to explain and act. The central concern is how to make useful causal claims about particular relationships while connecting them to the history that precedes and follows them.

Keywords: causal attribution; feedback; adaptive systems; causal separability; causal salience; intervention; trajectories; coevolution

1. Do the Times Make the Hero?

An old question can be stated in two opposite ways: Do the times make the hero, or does the hero make the times?

Imagine a leader who gains influence during a period of institutional strain. Established arrangements have become less effective, and people are looking for a different direction. Those conditions help explain why this person can become consequential. Once influential, however, the leader can redirect events, alter institutions, and change what others expect. The environment that made the leadership possible is subsequently changed by the leader's actions.

Research on historical institutional change examines this connection between the constraints actors inherit and the institutions their actions maintain or change [1]. The opening question brings that connection into ordinary causal language. We can reasonably emphasize either the conditions that gave an actor influence or the changes the actor then helped produce. What needs explaining is how these apparently opposite attributions relate to the same historical development.

We can slow down the hypothetical sequence. Let S denote the social environment and H the actor's position and capacity to act. The subscripts below mark successive stages: an environment S_0 helps create the conditions for an influential actor H_1 ; the actor contributes to a changed environment S_2 ; that environment changes what the actor can do at H_3 .

$$S_0 \rightarrow H_1 \rightarrow S_2 \rightarrow H_3 \rightarrow \dots$$

The arrows organize the sequence; they do not establish the causal contribution of each element. An observer concentrating on its first part may emphasize the times making the hero. Another concentrating on the next part may emphasize the hero making the times. Their disagreement may involve competing explanations, but it may also arise because they are explaining different stages of one history.

The same question recurs beyond leadership. Do institutions shape behavior, or does repeated behavior reproduce institutions? Does technology transform society, or do social needs and institutions shape technology? Does politics make policy, or does policy reshape politics? In each case, the objects being compared can continue interacting after the moment we choose to examine.

This essay follows that interaction. We begin with an ordinary cause-and-effect relationship, then ask what changes when its consequence becomes part of the next situation. That will allow us to distinguish the mechanisms operating in a system from the evidence available to identify them—and to examine why people may feel compelled to act before the distinction is clear.

2. The Comfortable World of $X \rightarrow Y$

Suppose a switch is flipped and a lamp turns on. The wiring, power supply, and bulb must be working, but under those conditions the basic causal picture is straightforward. An event occurs, and another follows.

Let $X(t)$ denote the switch being flipped at time t , and $Y(t + \Delta t)$ the lamp turning on after an elapsed time Δt . For a positive elapsed time, the sequence is:

$$X(t) \rightarrow Y(t + \Delta t), \quad \Delta t > 0$$

Time gives the events an ordering: the candidate cause occurs before the candidate consequence. That ordering is useful, although it does not by itself establish causation. A common cause could produce two successive events, a coincidence could resemble a causal sequence, or an omitted condition could change our explanation. Establishing the switch's effect requires evidence about the connection, not merely the order.

Within this picture, prediction and attribution ask different questions about the same sequence. Before the switch is flipped, we can ask whether the lamp will turn on. After the lamp turns on, we can ask what made it do so. Prediction looks forward from the available conditions; attribution looks backward from the observed event.

This simple picture supports much of everyday reasoning, scientific experimentation, policy analysis, and management. It is especially useful when we can define the question closely enough that the relevant relationship remains reasonably stable. In the lamp example, we can focus on what turning the switch does without explaining every subsequent event in the room.

The boundary of that question matters. If the light changes what the person can see, and what the person sees prompts another action, the first consequence has entered a further sequence. The original

explanation can remain correct while describing only one part of what follows. Public policy makes that continuation particularly visible.

3. Let the World Keep Moving

Politics helps determine which policies are enacted. Elections, coalitions, public opinion, institutions, interest groups, and leadership all influence the result. If our question concerns the enactment of a particular policy, politics is a natural place to look for its causes.

An enacted policy then changes the conditions under which subsequent politics occurs. It may distribute resources, create beneficiaries, alter administrative capacities, change expectations of government, or affect participation. Policy-feedback research examines how policies become inputs into later politics [2,3]. What was an outcome at one stage can become part of the explanation at the next.

Consider a hypothetical benefit program. The coalition that enacts it helps determine who is eligible and what recipients receive. Once the program operates, recipients may organize to protect it, administrators acquire responsibilities, and political candidates must decide how to respond. The program has become part of the environment in which a later coalition forms. Explaining its enactment and explaining its subsequent political influence are related tasks, but they concern different points in the sequence.

This is the basic structure of feedback: a change in one component affects another, whose response later affects the first. Politics can influence policy, policy can influence later politics, and the resulting politics can change the next policy. Cause and effect retain their meaning when we specify the relationship and stage to which they refer.

The same structure appears when market conditions shape firms whose choices then alter markets, or when a movement elevates leaders who subsequently redirect it. The current state can contain traces of these earlier interactions. Today's preferences may partly reflect yesterday's policies; today's institutions may embody earlier responses to problems.

Dynamic and cyclic causal models provide ways to study such relationships. Equilibrium causal models also show how long-run relations can sometimes be examined when the underlying dynamics contain feedback [4]. For our opening question, the practical implication is that looking at two quantities at one moment may leave out the sequence that connects them. The times may help make the hero at one stage, while the hero helps make later times at another.

So far, the examples suggest visible change. Feedback can also sustain a familiar pattern, however. To understand such a case, we need to ask what a stable state records and what ongoing activity it may conceal.

4. A Stable World Can Be Very Busy

Consider a social norm that remains broadly unchanged from one year to the next. People may follow it because they expect others to do so, anticipate approval or disapproval, or regard the behavior as normal.

Research on norm dynamics examines these expectations and the processes through which norms emerge, persist, and change [5,6].

Each act of compliance also enters the environment in which another person forms expectations. Seeing others follow the norm can encourage further compliance; resistance can weaken the expectation or prompt a response from those defending it. The norm influences behavior, while observed behavior helps sustain or alter the norm. A familiar annual pattern may therefore depend on continuous activity rather than the absence of change at every level.

This gives us a reason to distinguish stability in the observed state from simplicity in its underlying mechanism. People can be acting, observing, resisting, and adjusting even when the broad pattern moves very little. The question is what maintains that pattern.

Similar observations can also conceal different arrangements. Imagine one system with little pushing it in either direction and another with substantial opposing influences that nearly cancel. Both may show little net movement. Their responses to a disturbance could nevertheless differ, depending on the strength of those influences and how they adjust. The stable observation alone does not establish either system's resilience.

A closely divided political system offers an intuitive parallel. Pew Research Center found that U.S. registered voters were nearly evenly divided in partisan alignment in 2024 [7]. Research on affective polarization, meanwhile, documents substantial antagonism between partisans [8]. Numerical balance and internal agreement are different properties; observing the former gives us limited information about the latter.

The broader point is that a maintained state tells us where the system is, while explaining its maintenance requires evidence about what is happening within it. If different mechanisms can sustain the same state, what observation would help us distinguish them?

5. What the State Does Not Tell Us

A small thought experiment makes that question concrete. Suppose two systems contain variables X and Y , and both settle at a state where X and Y equal zero. In one system, X affects Y . In the other, Y affects X . Observing this settled state gives us the same values in both cases, even though the proposed mechanisms point in different directions.

Now suppose we can change X while leaving the other conditions of the comparison fixed. In a simple one-way system where X has a nonzero effect on Y , we would expect a response in Y . In the reverse one-way system, changing X alone would leave Y unchanged. Those contrasting responses could help distinguish the mechanisms that the original state left unresolved.

The comparison depends on its conditions: we have specified one-way relationships, a relevant change in X , and a way to isolate that change. Hidden common causes or additional feedback paths would require a more developed design. The experiment illustrates how a new observation can discriminate between particular explanations rather than supplying a general rule that settled states are uninformative. Equilibrium observations can support causal inference under suitable assumptions and designs [4].

We can call the degree to which available evidence distinguishes competing causal explanations causal separability (CS). The term is used here descriptively, rather than as a formal measure. Its value is to keep an evidentiary question separate from the mechanism itself: how well can these observations discriminate among the explanations we are considering?

A causal relationship can exist even when our observations leave it unresolved. Its effect can also be large without being easy to distinguish from competing causes. Conversely, a carefully designed comparison may identify a modest effect clearly. Existence, magnitude, and evidentiary discrimination are related concerns, but they answer different questions.

This distinction helps us assess an argument that moves from coexistence to explanation. If X and Y appear together in a stable relationship, we may be able to construct a story about X causing Y, Y causing X, or another factor influencing both. The useful next question is which evidence would choose among those stories. The response to a disturbance supplies one possibility, and following that response over time may supply more.

6. The Path Can Tell a Different Story

Return to the two systems, but suppose both now contain feedback between X and Y. They settle at the same state, and a disturbance to X produces the same first response. Our initial observation still leaves the mechanisms indistinguishable.

As the response travels through the system, later observations can differ. X changes Y, and Y feeds back into X. Differences in the strength or timing of that return path can produce different adjustment paths even if both systems eventually settle at the same state again. A comparison of the starting and ending values would miss the difference that appeared between them.

A hypothetical heating example makes this easier to picture. Two controllers maintain the same room temperature. After a heat disturbance, the first temperature readings look similar. One controller then restores the temperature gradually; the other produces an overshoot before settling. Both eventually reach the same baseline. The response path gives us a reason to investigate different control mechanisms, even though the maintained temperature and final reading match.

We are therefore examining three distinct observational objects: a state at one moment, a change between endpoints, and the trajectory connecting them. Each can carry information the others omit. Which is useful depends on the mechanisms we are trying to distinguish and on how we observe them.

The time horizon follows directly from this distinction. An observation made too soon may precede the feedback response that separates two mechanisms. Observing the adjustment path can reveal a difference; observing only the eventual settled state can miss it again. Temporal causal-discovery research likewise examines how sampling and subsampling affect what structure can be recovered from time-series data [9]. The hero-and-times question has a related vulnerability: an early slice can emphasize conditions shaping an actor, while a later slice emphasizes the actor reshaping those conditions.

Established models also describe a response to a persistent stimulus that eventually returns to its original baseline. This behavior is called perfect adaptation. Studying it shows why a lasting intervention may

have a transient response that an equilibrium observation conceals [10]. The path matters because the disappearance of an endpoint difference need not mean that nothing happened.

Interpreting a path still requires causal evidence. A hidden common driver can produce a temporal pattern resembling a direct effect, and a response may fit more than one mechanism. Controlled disturbances, comparisons, and assumptions about the system help determine what the trajectory establishes. Following the path enlarges the evidence available for discrimination; it does not make discrimination automatic.

This makes intervention an important tool: rather than only watching where the system sits, we change something and observe the response. In a social system, however, the response can include people changing how they behave under the new conditions. The intervention then becomes part of the mechanism we are trying to understand.

7. The Intervention That Changes the Answer

Suppose an organization observes that a measured activity is associated with better performance. It introduces a target for that activity, expecting the relationship to continue. Employees now have a new reason to increase the recorded measure. Some may improve the underlying work; others may redirect effort toward what is counted. The historical association can become a less reliable guide because the target changes what producing the measure involves.

This hypothetical example illustrates a problem extending beyond the values of two variables. An intervention can change people's expectations, strategies, timing, or decision rules, thereby changing the relationship used to predict its effect. Organizations may redesign processes, markets may reprice, and political coalitions may form around a new rule. A policy introduced from outside a local relationship can set off responses within the system.

The Lucas critique addresses this problem in economic policy evaluation. Behavioral relationships estimated under one policy regime need not remain invariant when a proposed policy changes the rules to which people respond [11]. The relevant question is whether the relationship supporting a prediction will continue to hold under the conditions the intervention creates.

Here it helps to distinguish assignment from response. A randomly assigned treatment can remain exogenous: its assignment is independent of the prior conditions that otherwise determine the response. People's subsequent choices arise within the affected system and belong among the treatment's consequences. Their adaptation may change the response mechanism while the assignment retains that independence.

Intervention can consequently do two kinds of work at once: provide evidence about a response and contribute to changes in the process producing that response. An experiment can estimate what happens under its specified conditions, including adaptation during the study. Applying that result elsewhere or later requires examining whether the conditions and responses remain relevant.

Other interventions alter subsequent conditions without changing decision rules in the same way. A successful safeguard may reduce the problem that originally made it necessary. A subsidy may change

entry decisions and prices. A regulation may encourage avoidance. Following each case forward lets us identify the particular response instead of treating all changes as one generic form of feedback.

We have so far separated observation from intervention: someone studies the system, then chooses an action. In many social settings, the interpretation already influences choices before a formal intervention occurs. People invest, prepare, or withdraw because of what they expect to happen. To see how explanations can enter the trajectory itself, consider the AI expansion underway in 2026.

8. The Observer Enters the System

AI offers a useful case because we can observe important changes while their longer-run consequences remain open. Investment, data-center construction, organizational adoption, and changing work practices are visible. Wachter and Wachter report that the five largest U.S. technology firms spent \$380 billion on capital expenditure in 2025 and were forecast to spend roughly twice that amount in 2026 [12]. U.S. Census evidence shows adoption across firms, business functions, and worker tasks, with substantial variation in its extent [13]. These observations establish activity; explaining what it will produce requires following the process further.

One account starts with capabilities. Better models make useful applications possible, firms invest in those applications, and adoption improves productivity. Executive survey evidence reports heterogeneous experiences and expectations of gains [14], while an empirical review finds productivity gains that are uneven and often insufficiently verified [15]. The account gives us a plausible route from capability to output, along with reasons to examine where each connection is realized.

Another account begins with expectations of future value. Those expectations attract capital, which finances infrastructure and model development. Valuations and visible investment can then reinforce confidence in the anticipated transition. In this account, investment records both a response to demonstrated usefulness and a commitment to benefits that have yet to materialize. The distinction matters when present spending is offered as evidence of eventual returns.

A third account emphasizes the adjustment required between capability and widespread benefit. Infrastructure, skills, complementary technologies, experimentation, and organizational redesign can delay or limit the payoff. Evidence on adoption and work organization documents uneven adjustment, while the International Energy Agency examines the energy requirements and constraints accompanying the data-center buildout [13,15,16]. A limited current productivity effect could therefore reflect incomplete adjustment rather than settle the question of longer-run potential.

These accounts can operate together. The same observed investment may respond to existing capability, anticipated returns, and infrastructure needs. Distinguishing their contributions requires evidence about adoption, realized benefits, financing, and the conditions under which investment continues. A later trajectory can separate interpretations that fit the same present observation.

The uncertainty differs from the difficulty of explaining a completed history. When studying an earlier technology or a historical leader, we know much of what followed. That knowledge can encourage us to compress an uncertain path into a clean sequence whose outcome appears more obvious than it was. In a

live case, we face the opposite limitation: observations that might distinguish competing explanations have yet to occur. Retrospective compression and an incomplete unfolding path create different problems for attribution.

AI also shows why observers belong inside the story. A firm expecting large productivity gains may invest sooner. A worker anticipating displacement may seek new skills, and an educator or policymaker may change plans. Forecasts of electricity demand may influence infrastructure commitments. These are possible behavioral channels through which beliefs about a future can contribute to producing or changing it. Work on reflexive anticipations examines this broader relationship between predictions and the responses they elicit [18].

We must then interpret later observations with those responses in view. Rising investment could partly reflect confidence generated by earlier forecasts; a policy response could change the employment pattern against which a forecast is judged. An explanation may therefore influence the future evidence offered in its support or against it.

Following this interaction gives causal inquiry a further task: distinguish what a forecast predicted from what responses to the forecast helped bring about. It also raises a practical question. AI attracts intense demand for explanations because decisions about work, investment, and public resources are at stake. How closely does that demand correspond to the evidence capable of choosing among explanations?

9. When the Demand for Explanation Outruns the Evidence

A close political contest makes the distinction especially clear. Imagine a two-option contest decided by simple majority, with 49.9% supporting one side and 50.1% the other. These are hypothetical numbers. A small shift can reverse the winner even if the underlying distribution of views changes very little.

We need to specify the institutional rule because numerical proximity alone does not determine the consequence. The United States, for example, selects its president through the Electoral College rather than the national popular vote. Pew Research Center's accounts of the 2024 election nevertheless provide evidence of close national division: partisan alignment was nearly balanced, and the presidential popular-vote margin was 1.5 percentage points [7,19]. Evidence of strong party loyalty and affective polarization also shows why a close aggregate need not imply similar political attachments [8,20].

The hypothetical majority contest illustrates how a small change in an underlying state can cross a threshold and produce a large change in control. We can describe the distribution as relatively stable while regarding the institutional outcome as highly sensitive. The two descriptions concern different objects, just as the maintained state and its underlying mechanism did earlier.

That sensitivity creates strong demand for causal explanation. People ask whether an outcome resulted from the economy, turnout, candidate quality, policy, media, or a particular event. In a close contest, several factors could each plausibly make a difference larger than the winning margin. Establishing such a contribution would require evidence; the small margin itself cannot identify one uniquely decisive cause.

We can call the attention and pressure directed toward a causal question causal salience (CA). As with causal separability, this is a descriptive term. It concerns how strongly people seek an explanation or feel

compelled to act on one. It does not measure a cause's objective importance. The political example is salient because the consequences of control make people care intensely about what produced the result.

Causal separability and causal salience therefore answer different questions. The first asks how well the evidence distinguishes competing explanations; the second asks how strongly people demand an explanation. A carefully controlled study may distinguish a relationship clearly while attracting little public interest. A crisis or close contest may attract extraordinary attention while leaving several explanations unresolved. Attention can rise faster than the evidence becomes discriminating.

This brings us back to AI. Its productivity, employment, valuation, and energy consequences create pressure to invest, reorganize, prepare, or regulate. The available evidence supports some observations more clearly than it supports their causal attribution. The ILO review reports uneven productivity gains and limited large-scale displacement so far [15]. The August 2026 revision of the Stanford payroll study reports a widening employment gap for young workers in highly AI-exposed occupations, alongside an absence of widespread aggregate displacement in the observed data [17]. The authors describe these as observational patterns rather than causal estimates of AI's effect.

That distinction gives the evidence a useful place in the discussion. The patterns identify differences to investigate; they leave open how much AI, labor-market conditions, education, hiring practices, and other developments contributed. Strong public concern does not resolve that attribution, although it can make investigation and timely decisions more urgent.

When pressure to act is strong and causal discrimination is limited, people still make consequential choices. They invest, regulate, vote, reorganize, imitate, or resist under uncertainty. Those responses enter the next situation and can change the relationships later evidence will describe. We can now return to the opening puzzle with a fuller question: how do the interacting components, and the ways they influence one another, change along the path?

10. Which Made Which?

Do the times make the hero, or does the hero make the times? We can now see why the question can support several different disagreements. Observers may be explaining different stages, using different time horizons, or working at different levels. They may also be treating an evolving relationship as if it remained fixed throughout the history.

Consider technology and an organization. A new technology changes what workers can do. The organization redesigns its workflow around those possibilities. The redesign creates new user needs, which influence what developers build next. At the next stage, the technology encounters an organization changed by its earlier use, and the organization encounters a technology partly shaped by that response.

Research on the social construction of technology examines how technological artifacts and relevant social groups shape one another [21]. Work on technological and institutional coevolution similarly studies mutually conditioning changes rather than treating either component as a passive recipient [22]. These perspectives help us ask which parts of an observed change arise from a specific influence and which arise from the continuing interaction.

A compact representation can make that distinction explicit. Let X_t denote the state of the technology and Y_t the state of the organization at stage t . Let f_t and g_t denote the transition relationships leading to their respective next states, with the stage's other conditions understood as part of those relationships. Then:

$$X_{t+1} = f_t(X_t, Y_t); \quad Y_{t+1} = g_t(X_t, Y_t)$$

The subscripts on X and Y indicate changing states; the subscripts on f and g indicate that the relationships themselves may change. The expressions are conceptual shorthand, not fitted equations or a complete model. In the example, changing workflows can alter how a given capability affects work, while changing user needs can alter how organizational experience affects development.

A directional question remains useful within this interaction. We can ask whether introducing a specified tool changed a particular practice under specified conditions. Extending that conclusion across stages requires examining what remained stable and what changed. The longer history may require both local causal estimates and an account of how the relevant conditions developed.

The same discipline applies to the hero-and-times question. Identifying a leader's contribution at one stage need not explain why that leadership became possible or how its influence later changed. A wider pattern can help us investigate those surrounding conditions. Scientific discovery provides a useful case because a familiar individual achievement sometimes appears in more than one place at nearly the same time.

11. What Does the Pattern Reveal?

Discoveries often become closely associated with individual names. The resulting account gives us a natural starting explanation: exceptional capability and work produced an exceptional idea. That explanation can be valuable while leaving a further question about why the discovery became possible when it did.

The history of science repeatedly includes multiple or near-simultaneous discoveries. Merton's studies examine such "multiples" as a recurring pattern [23,24]. When independent investigators approach a similar result, the pattern directs attention toward conditions they may share. Individual capability remains part of the inquiry, alongside the field's accumulated knowledge, available tools, and focused problems.

Merton identifies prerequisite knowledge and tools, focused attention, and scientific or social needs as relevant conditions [24]. Several investigators' progress can therefore provide evidence about the environment in which they worked. We still need to examine the particular cases: apparent independence, shared sources, and differences among results matter. The pattern gives us a reason to investigate common conditions rather than establish that the discovery was inevitable.

This returns us to the relation between an actor and the times. Asking who made a discovery concerns the work that produced a particular result. Asking why similar discoveries appeared together concerns the setting that supported those efforts. The two questions can inform one another when we preserve their respective objects of explanation.

Observable patterns can thus help us learn about parts of a system that are less directly visible. A cluster of innovations or similar organizational responses may suggest shared opportunities, constraints, or pressures. Such an inference depends on evidence connecting the pattern to those proposed conditions; resemblance alone does not identify the mechanism.

Our inquiry has widened without losing the original causal task. We can study an actor's contribution, the conditions under which it became possible, and the subsequent interaction. Those questions give the opening puzzle a more useful structure than choosing one permanent direction of influence.

12. A Puzzling System

Do the times make the hero, or does the hero make the times? A useful answer begins by specifying what we want to explain and where it occurs in the history. Conditions may help an actor become consequential; the actor may then change those conditions. Evidence about one stage can support an important attribution while leaving the other stage to be investigated.

Following the sequence also tells us what observations to seek. A stable state may be maintained by active feedback, and several mechanisms may sustain the same visible result. Response paths can reveal differences that endpoints conceal. Interventions can help distinguish explanations, while the responses they elicit become part of the subsequent system.

In a live social setting, even explanations can enter that sequence. People act on forecasts and attributions, affecting the events they later observe. The demand for an explanation can therefore intensify before the evidence discriminates well among alternatives. Keeping causal salience distinct from causal separability helps us recognize both the practical need to decide and the limits of the account guiding the decision.

The ordinary causal arrow remains useful throughout. Its task is to identify a local relationship that can then be connected to what preceded and followed it. Experiments, comparisons, counterfactual reasoning, historical analysis, and explicit models help establish that relationship. We should also ask whether it continues to hold at the next stage.

Following the evolving system allows us to identify contributions, distinguish alternatives, and learn from patterns while keeping their place in the history clear. A causal claim becomes more useful when readers can see both what it explains and how it connects to what comes next.

The world may be puzzling not because it lacks causes, but because causes do not wait patiently for us to finish attributing them before becoming consequences, feedback, beliefs, interventions, and causes again.

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References

- [1] Thelen, K. (1999). Historical Institutionalism in Comparative Politics. *Annual Review of Political Science*, 2, 369–404. DOI: 10.1146/annurev.polisci.2.1.369.
- [2] Pierson, P. (1993). When Effect Becomes Cause: Policy Feedback and Political Change. *World Politics*, 45(4), 595–628. DOI: 10.2307/2950710.
- [3] Campbell, A. L. (2012). Policy Makes Mass Politics. *Annual Review of Political Science*, 15, 333–351. DOI: 10.1146/annurev-polisci-012610-135202.
- [4] Ryan, O., & Dablander, F. (2025). Equilibrium Causal Models: Connecting Dynamical Systems Modeling and Cross-Sectional Data Analysis. *Multivariate Behavioral Research*, 60(6), 1116–1150. DOI: 10.1080/00273171.2025.2522733.
- [5] Young, H. P. (2015). The Evolution of Social Norms. *Annual Review of Economics*, 7, 359–387. DOI: 10.1146/annurev-economics-080614-115322.
- [6] Gelfand, M. J., Gavrillets, S., & Nunn, N. (2024). Norm Dynamics: Interdisciplinary Perspectives on Social Norm Emergence, Persistence, and Change. *Annual Review of Psychology*, 75, 341–378. DOI: 10.1146/annurev-psych-033020-013319.
- [7] Pew Research Center. (2024). The Partisanship and Ideology of American Voters. April 9, 2024.
- [8] Iyengar, S., Lelkes, Y., Levendusky, M., Malhotra, N., & Westwood, S. J. (2019). The Origins and Consequences of Affective Polarization in the United States. *Annual Review of Political Science*, 22, 129–146. DOI: 10.1146/annurev-polisci-051117-073034.
- [9] Gong, C., Zhang, C., Yao, D., Bi, J., Li, W., & Xu, Y. (2025). Causal Discovery from Temporal Data: An Overview and New Perspectives. *ACM Computing Surveys*, 57(4), Article 100. DOI: 10.1145/3705297.
- [10] Blom, T., & Mooij, J. M. (2023). Causality and Independence in Perfectly Adapted Dynamical Systems. *Journal of Causal Inference*, 11(1), 20210005. DOI: 10.1515/jci-2021-0005.
- [11] Lucas, R. E., Jr. (1976). Econometric Policy Evaluation: A Critique. *Carnegie-Rochester Conference Series on Public Policy*, 1, 19–46. DOI: 10.1016/S0167-2231(76)80003-6.
- [12] Wachter, J. A., & Wachter, J. (2026). What Investment Data Implies about the AI Transition. NBER Working Paper 35290. DOI: 10.3386/w35290.
- [13] Bonney, K., Breaux, C. L., Dinlersoz, E., Foster, L. S., Haltiwanger, J. C., & Pande, A. A. (2026). The Microstructure of AI Diffusion: Evidence from Firms, Business Functions, and Worker Tasks. NBER Working Paper 35141. DOI: 10.3386/w35141.
- [14] Baslandze, S., Edwards, Z., Graham, J., McClure, T., Meyer, B. H., Sparks, M., Waddell, S. R., & Weitz, D. (2026). Artificial Intelligence, Productivity, and the Workforce: Evidence from Corporate Executives. NBER Working Paper 34984. DOI: 10.3386/w34984.
- [15] Merola, R., Ernst, E., Samaan, D., del Rio-Chanona, M., & Teutloff, O. (2026). The Impact of GenAI on Jobs, Productivity and Work Organization: A Review of the Empirical Evidence. International Labour Organization. DOI: 10.54394/00034628.
- [16] International Energy Agency. (2026). Key Questions on Energy and AI. IEA, Paris.

- [17] Brynjolfsson, E., Chandar, B., & Chen, R. (2025; revised August 12, 2026). Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence. Stanford Digital Economy Lab.
- [18] Patomäki, H. O. (2020). Reflexivity of Anticipations in Economics and Political Economy. In R. Poli (Ed.), *Handbook of Anticipation*. Springer. DOI: 10.1007/978-3-319-31737-3_16-1.
- [19] Pew Research Center. (2025). How Changes in Turnout and Vote Choice Powered Trump's Victory in 2024. June 26, 2025.
- [20] Pew Research Center. (2025). Voting Patterns in the 2024 Election. June 26, 2025.
- [21] Pinch, T. J., & Bijker, W. E. (1984). The Social Construction of Facts and Artefacts: Or How the Sociology of Science and the Sociology of Technology Might Benefit Each Other. *Social Studies of Science*, 14(3), 399–441. DOI: 10.1177/030631284014003004.
- [22] Zhang, Y. (2020). The Map Is Not the Territory: Coevolution of Technology and Institution for a Sustainable Future. *Current Opinion in Environmental Sustainability*, 45, 56–68. DOI: 10.1016/j.cosust.2020.08.017.
- [23] Merton, R. K. (1961). Singletons and Multiples in Scientific Discovery: A Chapter in the Sociology of Science. *Proceedings of the American Philosophical Society*, 105(5), 470–486.
- [24] Merton, R. K. (1963). Resistance to the Systematic Study of Multiple Discoveries in Science. *European Journal of Sociology / Archives Européennes de Sociologie*, 4(2), 237–282.